

CHANNEL DESIGN BASIC PRINCIPLES – ALLOWABLE VELOCITY METHOD

**Source of Information (on learn):
USDA-NRCS Part 654 Stream Restoration
Design, National Engineering Handbook –
Threshold and Alluvial channel design**

Channel Types

- ***Threshold Channels***

- “**Threshold**” applied forces from the design flow are below the threshold for movement of the boundary material (i.e. stable channel)
- Do not have the ability to adjust their geometry
- Morphological concepts typically do not apply
- Design methods well established:
 - allowable velocity or
 - allowable shear stress
- Need to evaluate sediment deposition (depending on sediment supply), especially at flows below the design flow
- Fine sediment passes through as “throughput” or “washload”. It is not considered part of bed-material or sediment load, but can be environmental issue

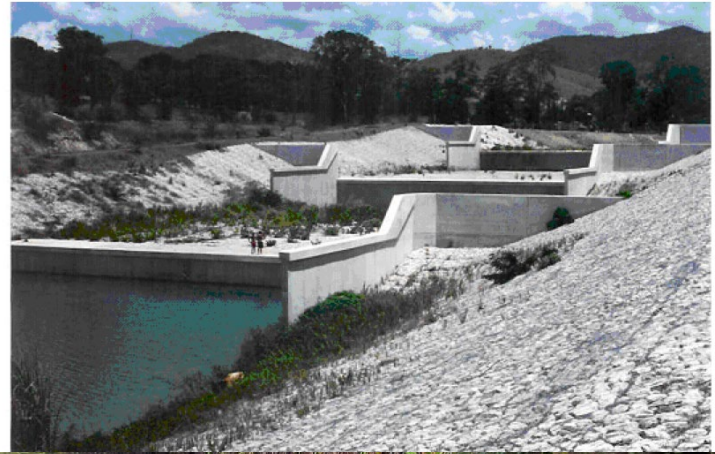




Figure 7-2 Grass-lined threshold channel

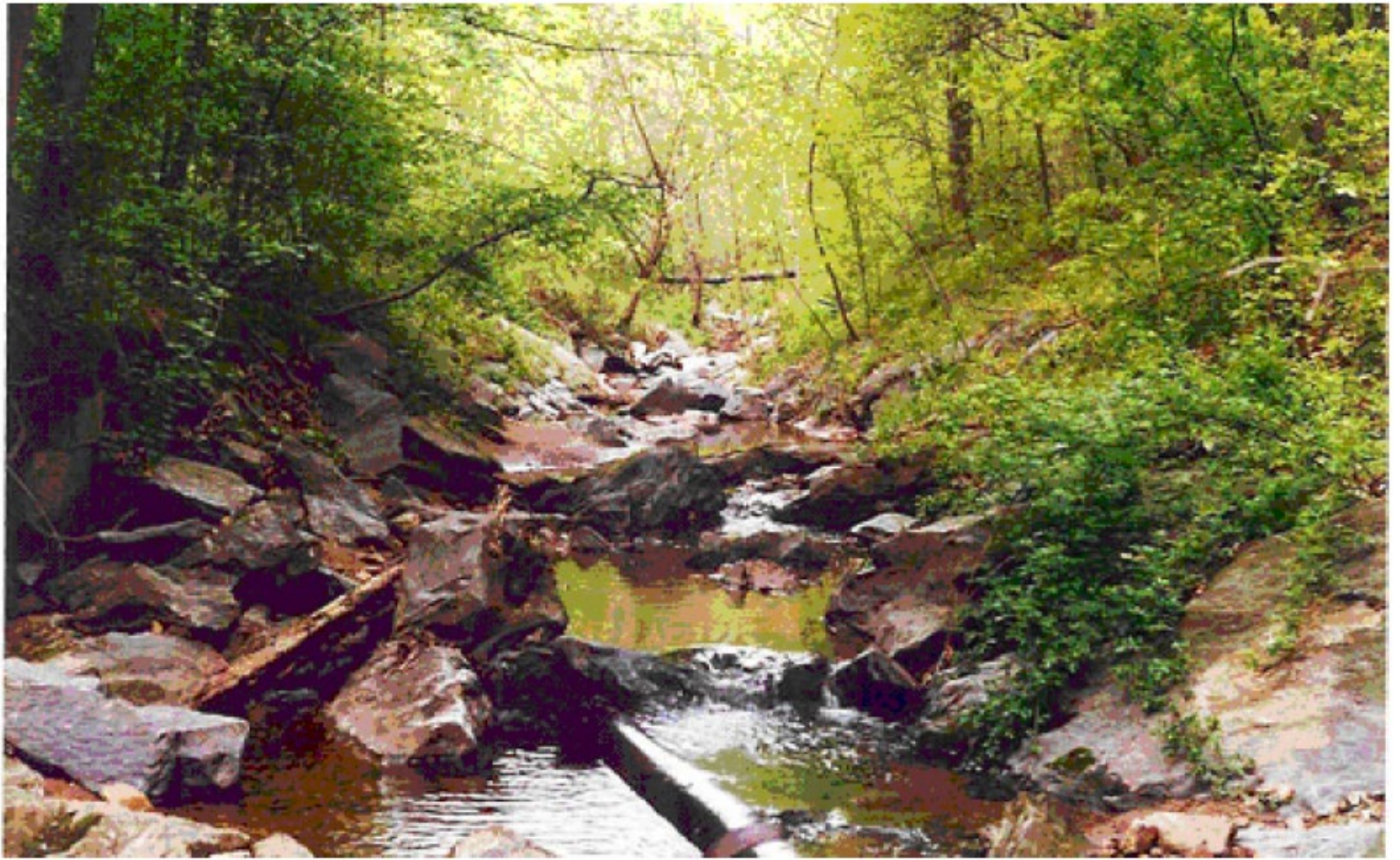


Figure 7-3. Bedrock threshold channel

Channel Types

- ***Alluvial Channels***

- Bed and banks formed of material transported by the stream under present flow conditions
- Adjust width, depth, slope and planform in response to changes in water or sediment
- Bank erosion and bankline migration are natural processes
- Stability: ability to pass the incoming sediment load without significant aggregation. Width, depth and slope are fairly consistent over time.
- Sediment transport analysis required for design
- Design needs to include application of geomorphic principles



Figure 7-4. Sand-bed alluvial channel

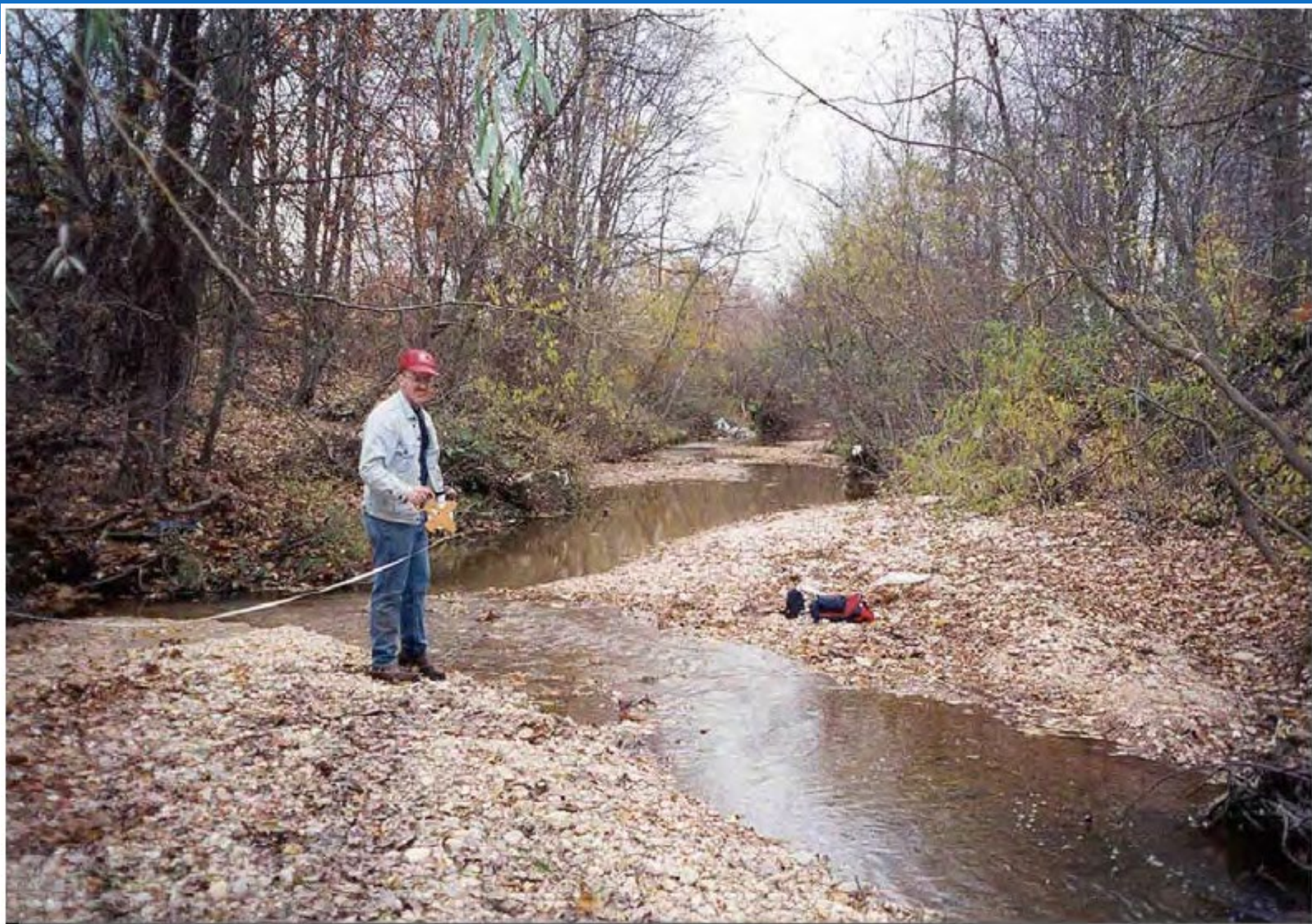


Figure 7-5. Sand- and gravel-bed alluvial channel



Figure 7-6. Gravel-bed alluvial channel



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Channel Types

- ***Transition Channels***

- Not always a clear distinction between threshold and alluvial channels
- Armouring: threshold at low flows and alluvial at high flows
- One reach may be threshold while another reach may be alluvial



Figure 7-7. Boulder-bed channel that could be a threshold channel or an alluvial channel, depending on the design discharge.



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Figure 7-8 Subsurface layer exposed after removal of three cobbles from the armour layer

Stream Flow Conditions

- **Perennial stream:** almost always has flow
 - A perennial stream is a stream that exhibits a measurable surface discharge more than 80 percent of the time
- **Intermittent stream:** flows only seasonally or sporadically, typically seasonal in nature
- **Ephemeral stream:** generally only flows after a significant rainfall event
- Channel process and morphology are significantly affected when discharge is intermittent (i.e. intermittent or ephemeral)



Ephemeral stream

Channel Design Variables

	Threshold Channel	Alluvial Channel
Channel boundary	Immobile at design discharge	Mobile
Bed-material sediment inflow	Usually small or negligible	Significant
Dependent variables	Width Depth Slope Roughness if there is a choice of boundary materials	Width Depth Slope Planform Bank roughness Roughness due to obstructions or structures
Independent variables	Design discharge Channel roughness	Design hydrograph Channel-forming discharge Bed-material sediment inflow Bed-material Streambank characteristics
Design equations	Energy Momentum Resistance	Energy Momentum Resistance Sediment transport Geomorphic relationship
Design goal with respect to channel stability	Pass the design discharge below the top of bank without mobilizing the boundary	Pass the incoming sediment load without significant aggradation or degradation or planform change.

Channel Design Methods and Approaches

	Threshold Channels	Alluvial Channels
Design Discharges	Maximum design discharge	Channel-forming discharge Flow-duration curve and/or long-term hydrograph
Design Criteria	Critical velocity/shear stress	Continuity of Sediment
Dependent Variables	Width, depth and slope, (roughness if there is a choice of boundary material)	Width, depth, slope, planform, bank roughness and roughness due to obstructions or structures
Design Equations	Energy, momentum and hydraulic resistance	Energy, momentum, hydraulic resistance, sediment transport and geomorphic relationship

Threshold Channel Design

Table 8-1: General guidance for selecting the most appropriate channel design technique

Technique	Significant sediment load and movable channel boundaries	Boundary material smaller than sand size	Boundary material larger than sand size	Boundary material does not act as discrete particles	No base flow in channel. Climate can support permanent vegetation
Allowable Velocity		X			
Allowable Shear Stress			X		
Tractive Power				X	
Grass Lined / tractive stress					X
Alluvial Channel Design Techniques	X				

Determine Allowable Velocity

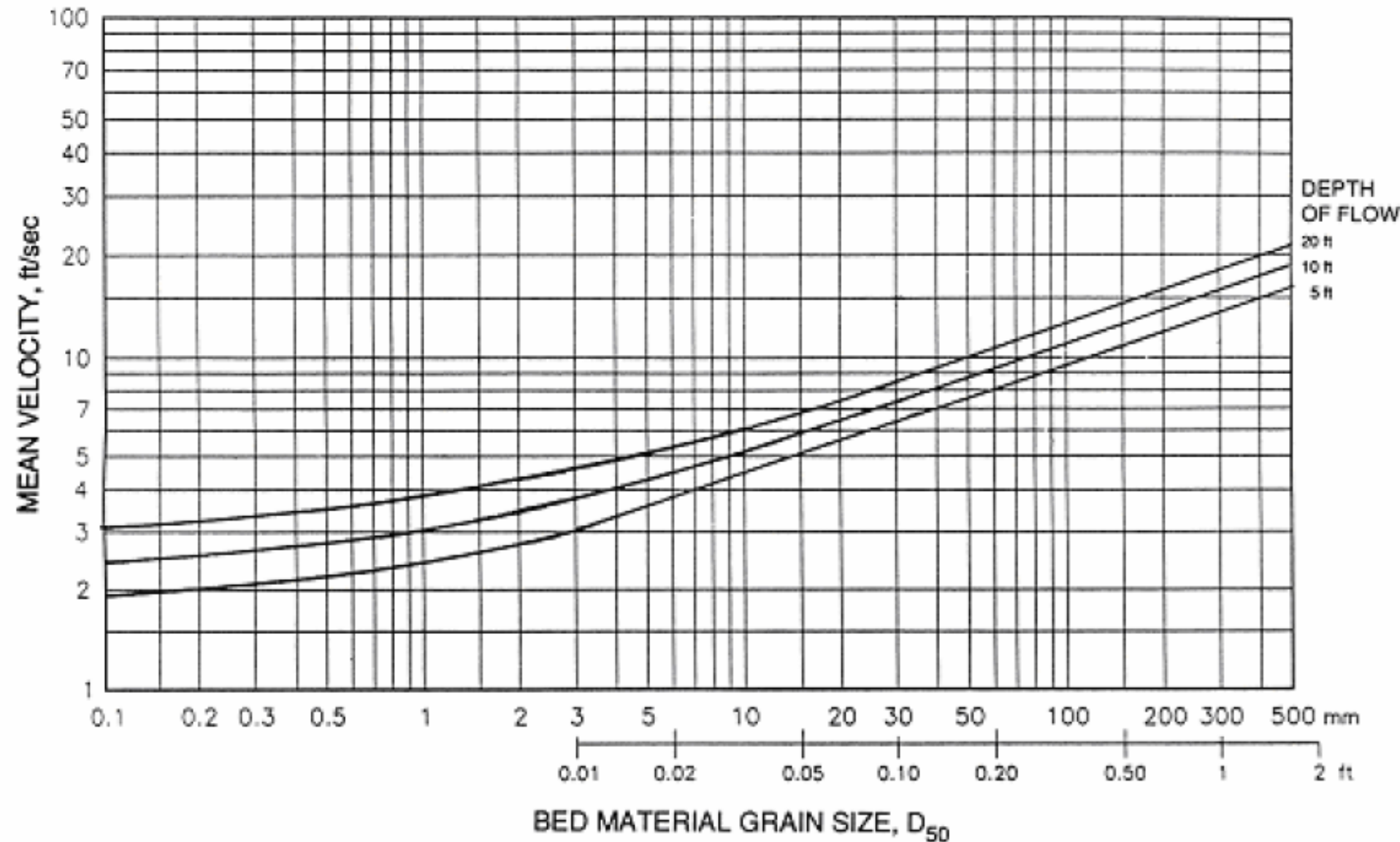
Table 8-3. Maximum Permissible Canal Velocities (Fortier and Scobey, 1926)

Original material excavated for canals	Mean velocity, for straight canals of small slope, after aging with flow depths less than 3 ft (0.9m)					
	Clear water, no detritus		Water transporting colloidal silts		Water transporting noncolloidal silts, sands, gravels or rock fragments	
	ft/sec	m/sec	ft/sec	m/sec	ft/sec	m/sec
Fine Sand noncolloidal	1.5	0.46	2.5	0.76	1.5	0.46
Sandy loam, noncolloidal	1.75	0.53	2.5	0.76	2.0	0.61
Silt loam, noncolloidal	2.0	0.61	3.0	0.91	2.0	0.61
Alluvial silt, noncolloidal	2.0	0.61	3.5	1.07	2.0	0.61
Ordinary firm loam	2.5	0.76	3.5	1.07	2.25	0.69
Volcanic ash	2.5	0.76	3.5	1.07	2.0	0.61
Stiff clay, very colloidal	3.75	1.14	5.0	1.52	3.0	0.91
Alluvial silt, colloidal	3.75	1.14	5.0	1.52	3.0	0.91
Shales and hardpans	6.0	1.83	6.0	1.83	5.0	1.52
Fine gravel	2.5	0.76	5.0	1.52	3.75	1.14
Graded, loam to cobbles when noncolloidal	3.75	1.14	5.0	1.52	5.0	1.52
Graded silt to cobbles when colloidal	4.0	1.22	5.5	1.68	5.0	1.52
Coarse gravel, noncolloidal	4.0	1.22	6.0	1.83	6.5	1.98
Cobbles and shingles	5.0	1.52	5.5	1.68	6.5	1.98

Table 8-4. Allowable Velocities (USACE 1994b)

Channel Material	Mean Channel Velocity	
	ft/sec	m/sec
Fine Sand	2.0	0.6
Coarse Sand	4.0	1.2
Fine Gravel	6.0	1.8
Earth		
Sandy Silt	2.0	0.6
Silt Clay	3.5	1.1
Clay	6.0	1.8
Grass-lined Earth (slopes less than 5 percent)		
Bermuda Grass		
Sandy Silt	6.0	1.8
Silt Clay	8.0	2.4
Kentucky Blue Grass		
Sandy Silt	5.0	1.5
Silt Clay	7.0	2.1
Poor Rock (usually sedimentary)	10.0	3.0
Soft Sandstone	8.0	2.4
Soft Shale	3.5	1.1
Good Rock (usually igneous or hard metamorphic)	20.0	6.1

Allowable velocity – depth data for granular materials (USACE 1994a)



Grass-Lined Channels - *Allowable Velocity*

Table 8-6. Allowable Velocities for Channels Lined with Grass U.S. Soil Conservation Service 1954

Cover	Slope range Percent	Allowable velocity, ft/sec	
		Erosion- resistant soils	Easily eroded soils
Bermuda grass	0-5	8	6
	5-10	7	5
	>10	6	4
Buffalo grass, Kentucky bluegrass, smooth brome, blue grama	0-5	7	5
	5-10	6	4
	>10	5	3
Grass mixture	0-5	5	4
	5-10	4	3
	Not recommended on slopes greater than 10 percent		
Lepedeza sericea, weeping love grass, ischaemum (yellow bluestem), kudzu, alfalfa, crabgrass	0-5	3.5	2.5
	Not recommended on slopes greater than 5 percent, except for side slopes in a compound channel		
Annuals – used on mild slopes or as temporary protection until permanent covers are established, common lespedeza, Sudan grass	0-5	3.5	2.5
	Not recommended for slopes greater than 5 percent		

Procedure for Application of Allowable Velocity Method (NEH 2007)

1. **Determine the hydraulics of the system.** This includes hydrologic determinations as well as the stage-discharge relationships for the channel considered.
2. **Determine the soil properties** of the bed and banks of the design reach and of the channel upstream.
3. **Determine the concentration of the suspended sediment load entering the reach.** This is best accomplished by measurements. Channels with suspended sediment concentrations less than 1,000 ppm are considered *sediment free*. Sediment free flows are considered to have no effect on channel stability. Channels with suspended sediment concentrations greater than 20,000 ppm are considered to be *sediment laden*. Sediment laden flows are considered to enhance sediment stability by filling boundary interstices with cohesive material. If a significant portion of the inflowing sediment load is bed-material load it is likely that the channel is alluvial and threshold design methods are not applicable.

Procedure for Application of Allowable Velocity Method (NEH 2007)

4. **Check to see if the allowable velocity procedure is applicable** using the Channel Evaluation Procedural Guide, Table 8-1.
5. **Determine the basic average allowable velocities for the channel** from one or more of the available design guidelines (Table 8-3, Table 8-4, and Figure 8-4). Multiply the basic allowable velocity by the appropriate correction factors (Figure 8-4).
6. **Compare the design velocities with the allowable velocities.** If the allowable velocities are greater than the design velocities, the design is satisfactory. Otherwise, one of the three options are available:
 - a. Redesign the channel to reduce velocity,
 - b. Provide structural measures (riprap, grade control, etc.) to prevent erosion, or
 - c. Consider a mobile boundary condition and evaluate the channel using appropriate sediment transport theory and programs.

Velocity Calculations

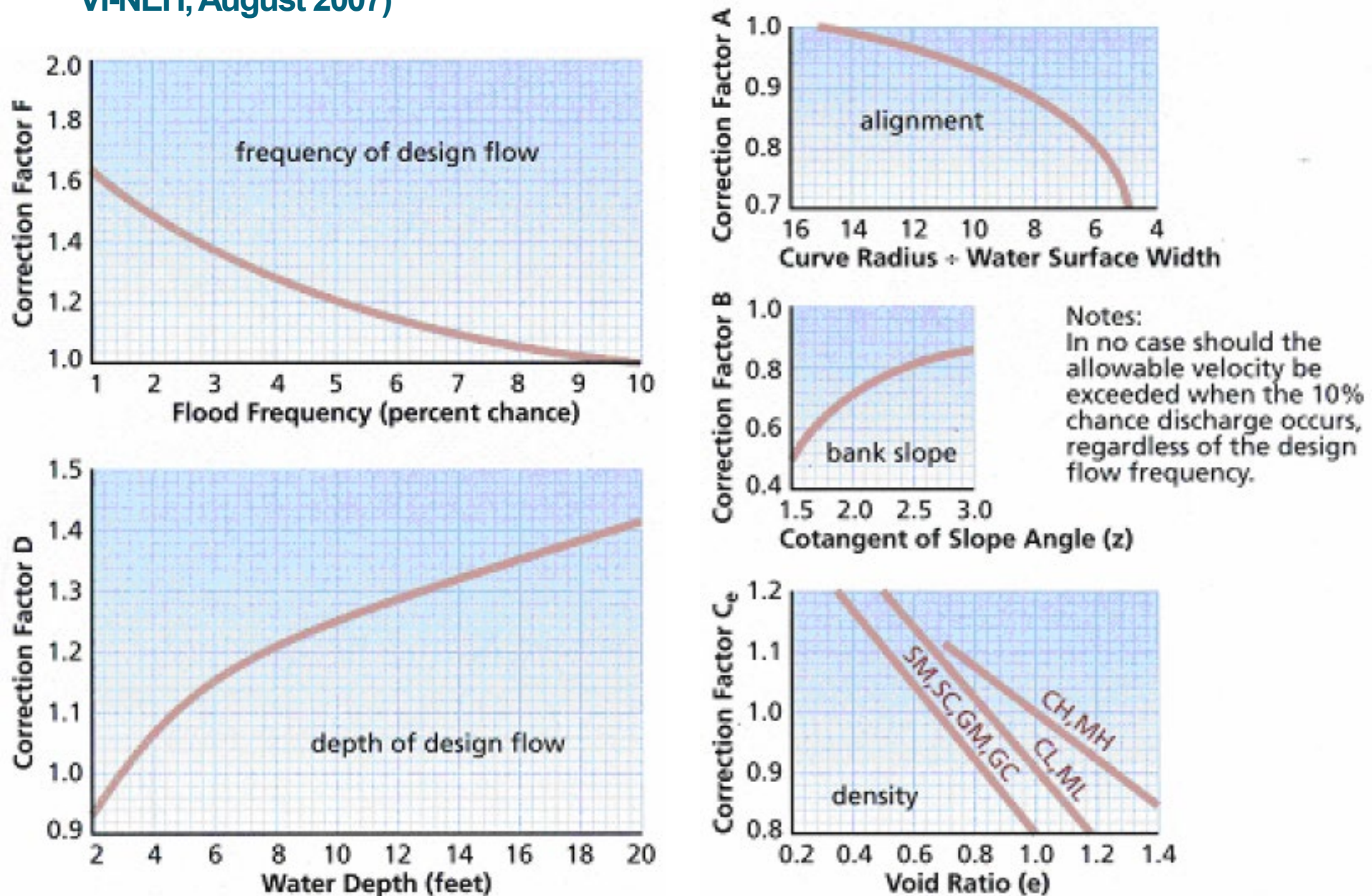
- Manning's Equation (Chezy Equation)

$$V = \frac{1}{n} R^{2/3} S^{1/2}$$

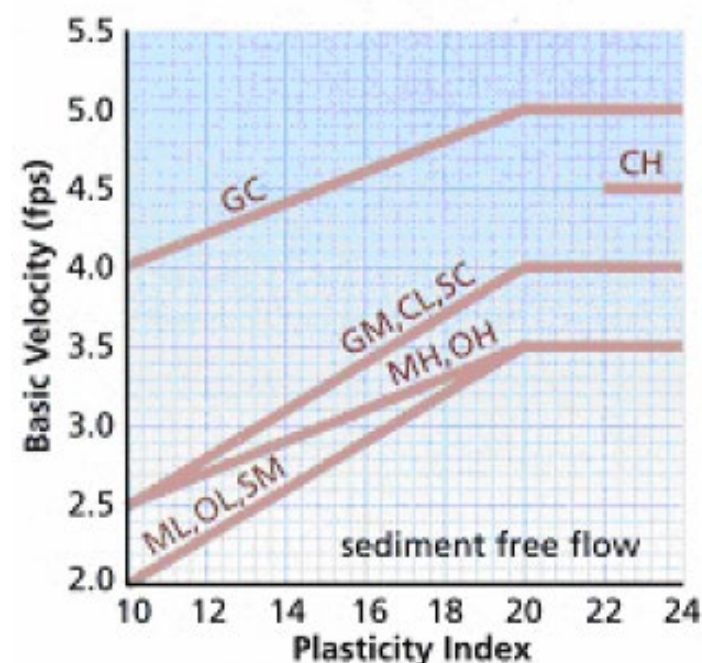
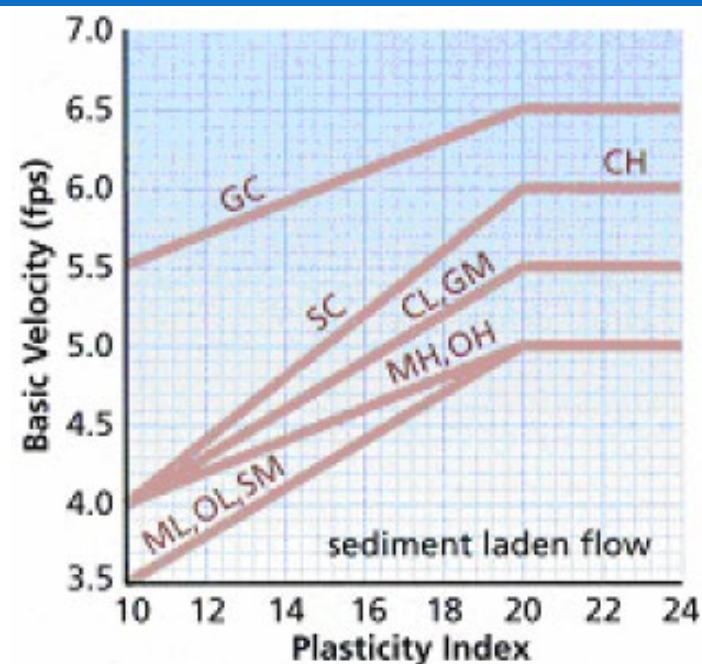
$$Q = \frac{1}{n} A R^{2/3} S^{1/2}$$

- Hydraulic Models such as HEC RAS

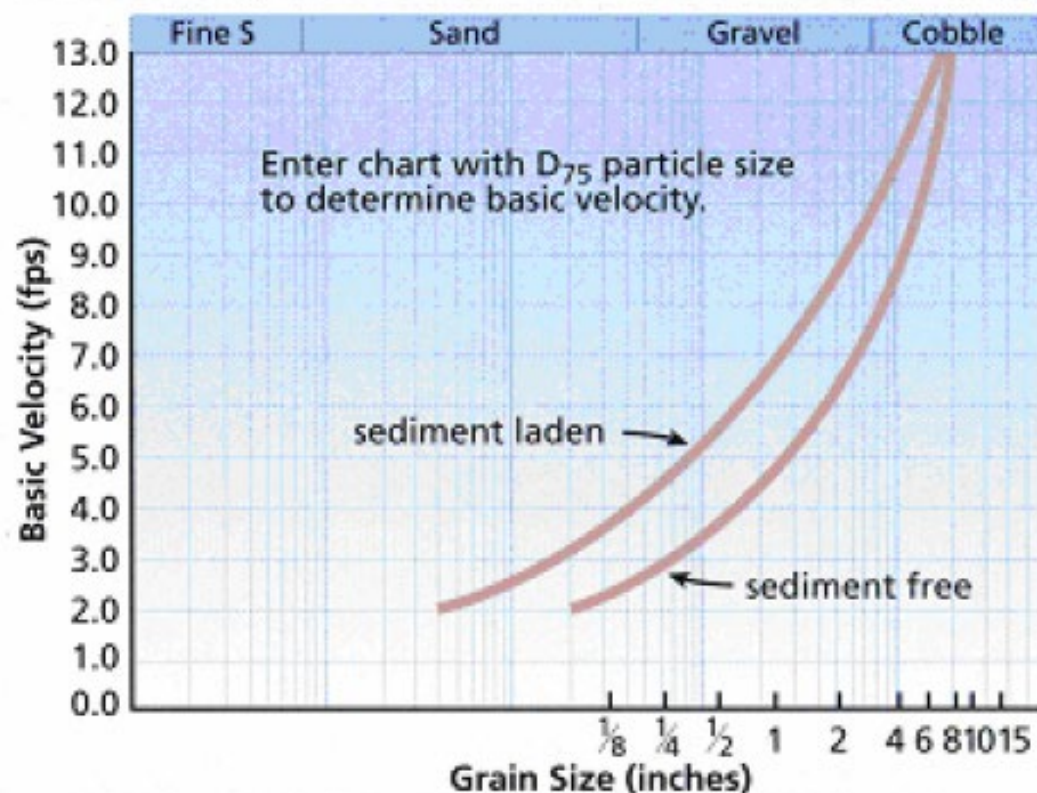
Figure 8.4 Allowable velocities for unprotected earth channels (USDA 1977 and 210-VI-NEH, August 2007)



Basic Velocities for Coherent Earth Materials (v_b)



Basic Velocity for Discrete Particles of Earth Materials (v_b)



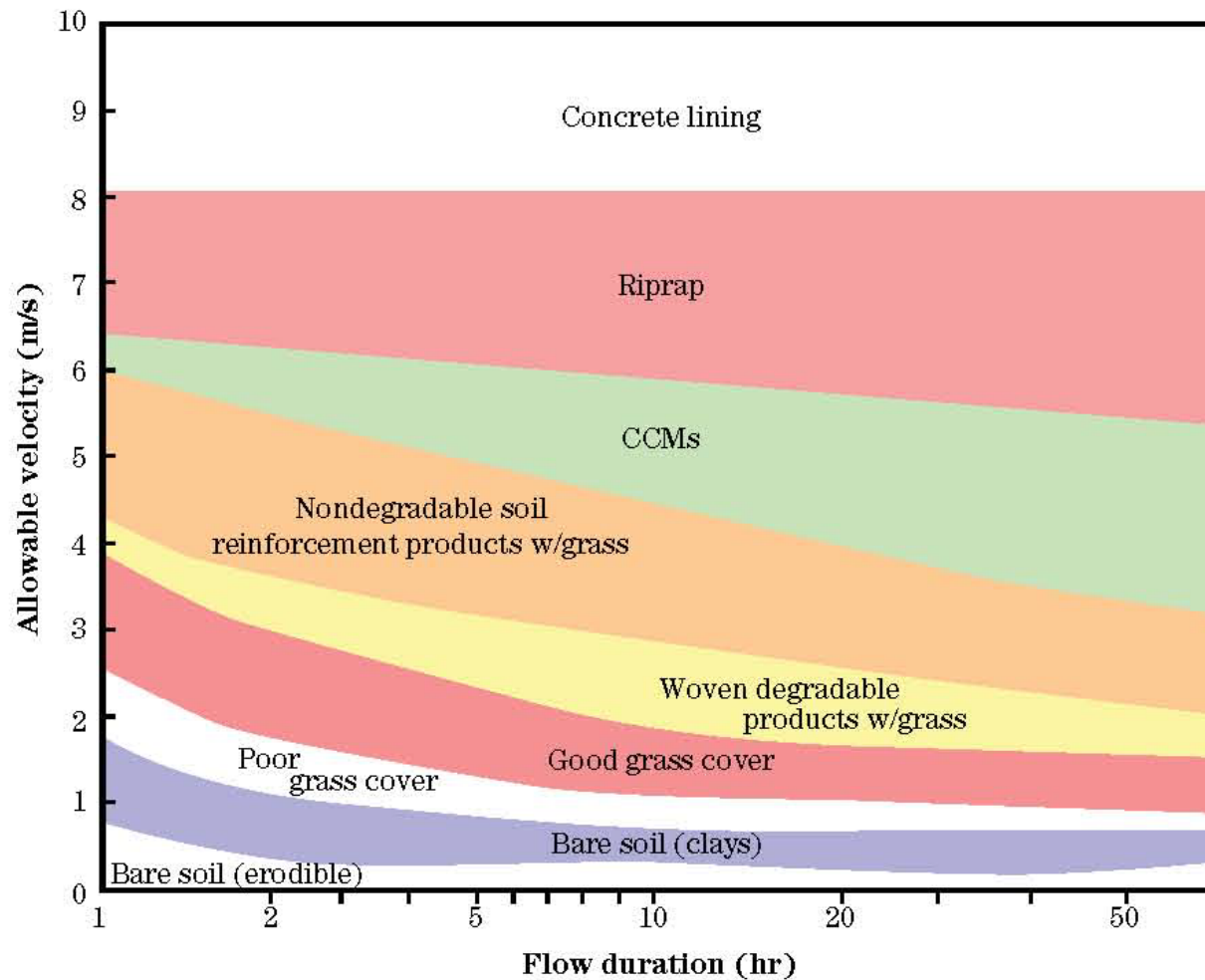
Allowable Velocities for Unprotected Earth Channels	
Channel Boundary Materials	Allowable Velocity
Discrete Particles	
Sediment Laden Flow	
$D_{75} > 0.4\text{mm}$	basic velocity chart value x D x A x B
$D_{75} < 0.4\text{mm}$	2.0 fps
Sediment Free Flow	
$D_{75} > 0.2\text{mm}$	basic velocity chart value x D x A x B
$D_{75} < 0.2\text{mm}$	2.0 fps
Coherent Earth Materials	
PI > 10	basic velocity chart value x D x A x F x C_e
PI < 10	2.0 fps

Allowable Velocity and Shear for Channel Lining Materials

Boundary Category	Boundary Type	Allowable Velocity ft/sec	Allowable Shear Stress lb/sq ft	Citation(s)
Temporary Degradable RECPs	Jute net	1 -2.5	0.45	E,H,M
	Straw with net	1 -3	1.5 – 1.65	E,H,M
	Coconut fiber with net	3 -4	2.25	E,M
	Fiberglass roving	2.5 -7	2	E,H,M
Non-Degradable RECPs	Unvegetated	5 -7	3	E,G,M
	Partially established	7.5 -15	4 – 6	E,G,M
	Fully vegetated	8 -21	8	F,L,M
Soil Bioengineering	Wattles	3	0.2 – 1.0	C,I,J,N
	Reed fascine	5	0.6 – 1.25	E
	Coir roll	8	3 – 5	E,M,N
	Vegetated coir mat	9.5	4 – 8	E,M,N
	Live brush mattresses (initial)	4	0.4 – 4.1	B,E,I
	Live brush mattresses (grown)	12	3.9 – 8.2	B,C,E,I,N
	Brush layering (initial/grown)	12	0.4 – 6.25	E,I,N
	Live fascine	6 -8	1.25 – 3.1	C,E,I,J
	Live willow stakes	3 -10	2.1 -3.1	E,N,O
Hard Surface	Gabions	14 -19	10	D
	Concrete	> 18	12.5	H
¹ Ranges of values generally reflect multiple sources of data or different testing conditions				
B. Florineth (1982)	F. Julien, P.Y. (1995)	J. Schoklitsch, A. (1937)		
C. Gerstgrasser, C. (1998)	G. Kouwen, N et al. (1980)	M. TXDOT (1999)		
D. Groff, K. (1999)	H. Norman, J.N. (1975)	N. Fischenich (2001)		
E. Gray, D.H. and Sotir, R.B. (1996)	I. Schiechl, H.M. and Stern, R. (1996)	O. USACE (1997)		

Duration of Design Flow (velocity)

Figure 8-25 Effect of flow duration on allowable velocities for various channel linings



*For slopes <5%